Mixed-effects Regression Models for Continuous Outcomes

Eliot Weinstein
eliotw@uchicago.edu
and
erweinstein.com

Zoom Office Hours

Friday and Saturday afternoons:

https://calendly.com/eliotw-uchicago/30min

In-person Office Hours ***Tuesday afternoons***
Tentatively, 4:00-6:00pm Hutch/Reynolds Club

-->But feel free to email me if none of those options work for you.

eliotw@uchicago.edu

Thanks!

Mixed-effects regression models (MRMs)

- Random-effects models
- Variance component models
- Multilevel models
- Hierarchical linear models
- Two-stage models
- Random coefficient models
- Mixed models
- Empirical Bayes models
- Random regression models

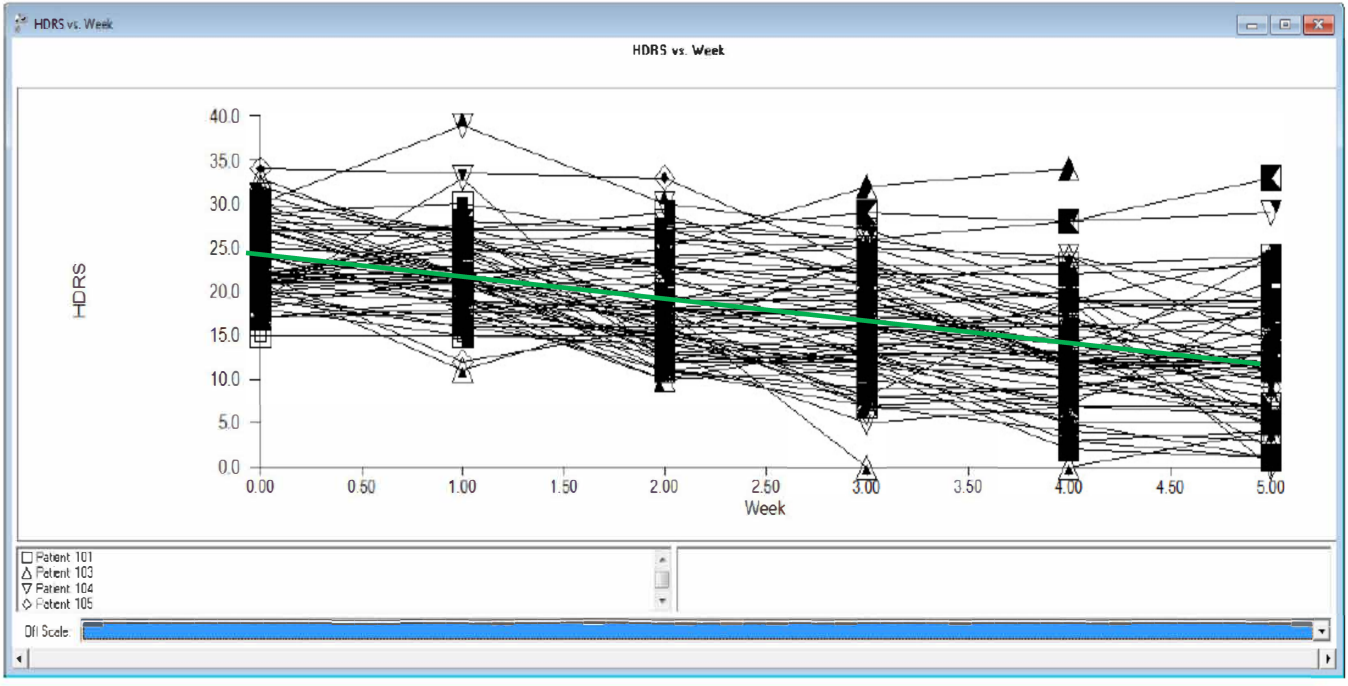
Basic characteristics of the models

- Including random effects that capture the correlational structure of the data
- Outcomes are correlated, examples:
  - Correlation ***within subjects*** for repeated observations
  - Clustering effects ***within strata*** (e.g. school, hospital, county etc)

A simple linear regression model

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + \varepsilon_{ij}$$

- Subscripts: ***individual i*** (i=1, 2, 3,…) at ***occasion j*** (j=1, 2, 3, …)
- Outcome Y
- Independent variable t
- Error term ε_{ij}
  - Assumption?
  - "iid", independent and identically distributed
  - $\varepsilon_{ij} \sim NID\ (0, \sigma^2)$


HDRS vs. Week
HDRS vs. Week
HDRS
40.0
35.0
30.0
25.0
20.0
15.0
10.0
5.0
0.0
0.00
0.50
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1.50
2.00
2.50
3.00
3.50
4.00
4.50
5.00
Week
Patient 101
Patient 103
Patient 104
Patient 105
Dft Scale:

Random intercept MRM

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + \boldsymbol{v_{0i}} + \varepsilon_{ij}$$

A model with a hierarchical or multilevel form

Random intercept MRM

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} \mathbf{+ \boldsymbol{v}_{0i}} + \varepsilon_{ij}$$

A hierarchical or multi**level** form -- OK, so what are the levels?

- Within-subjects (level 1): each unit/subject has multiple observations

- Between-subjects (level 2): the level 1 unit/subject records nest within level 2 entities

Random intercept MRM

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + \boldsymbol{v_{0i}} + \varepsilon_{ij}$$

A hierarchical or multilevel form

- Within-subject (level-1) model:

$$Y_{ij} = b_{0i} + b_{1i} t_{ij} + \varepsilon_{ij}$$

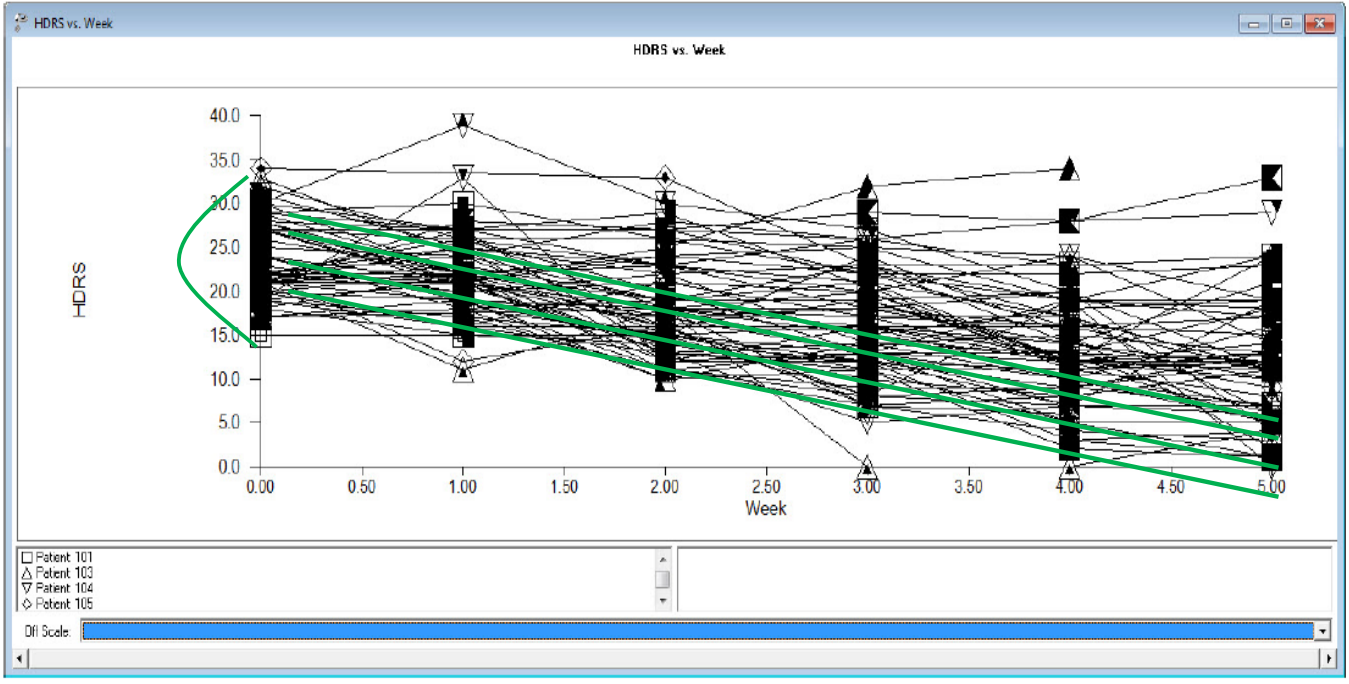
$$\varepsilon_{ij} \sim N\,(0, \sigma^2)$$

- Between-subject (level-2) model:

$$b_{0i} = \beta_0 + \boldsymbol{v_{0i}}$$

$$b_{1i} = \beta_1$$

$$v_{0i} \sim N\,(0, \sigma_v^2)$$


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4.00
4.50
5.00
Week
Patient 101
Patient 103
Patient 104
Patient 105
Dfl Scale:

Random intercept **and trend** MRM

The hierarchical or multilevel form

- Within-subject (level-1) model:

$$Y_{ij} = b_{0i} + b_{1i}t_{ij} + \varepsilon_{ij}$$

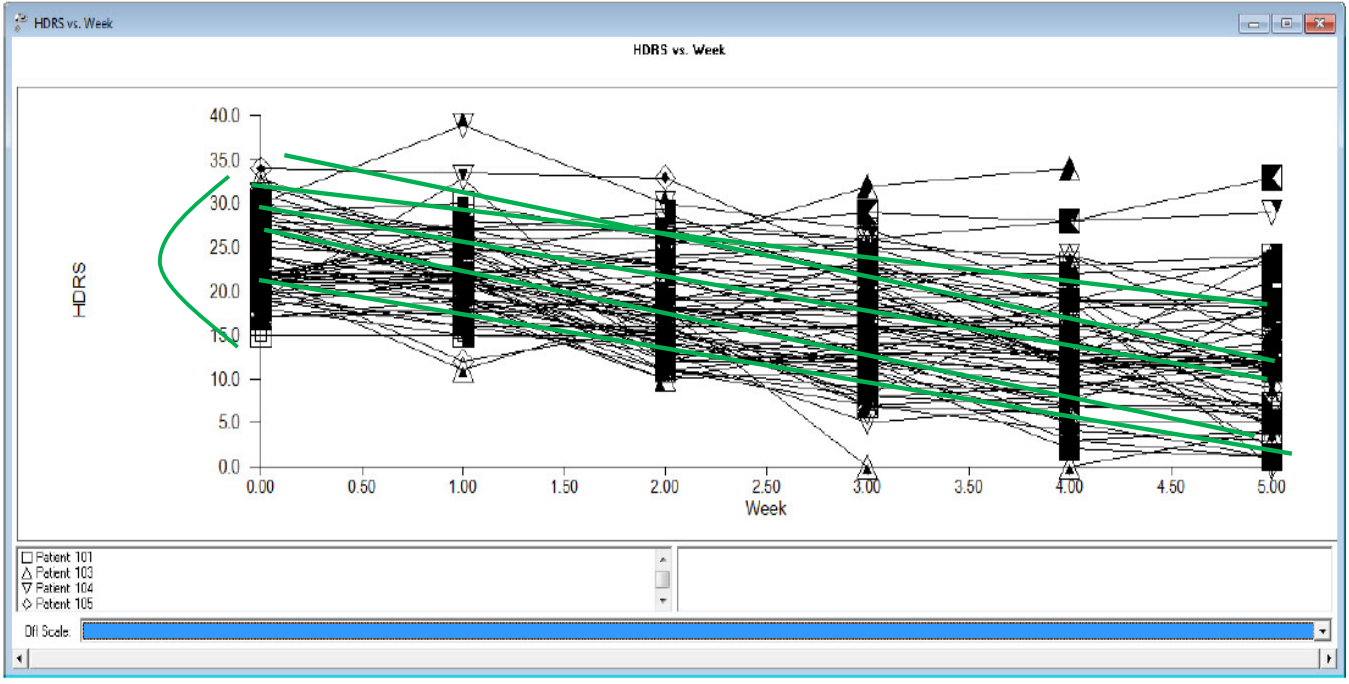
$$\varepsilon_{ij} \sim N\,(0, \sigma^2)$$

- Between-subject (level-2) model:

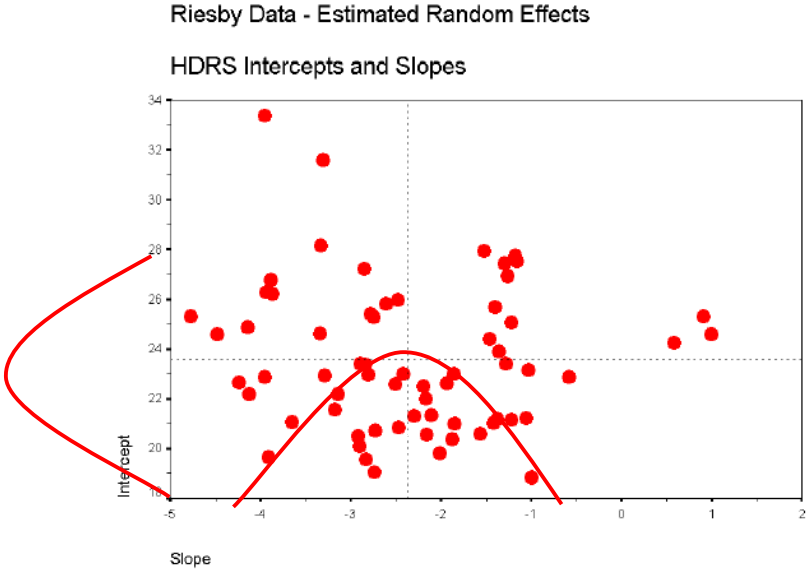
$$b_{0i} = \beta_0 + v_{0i}$$

$$b_{1i} = \beta_1 + \boldsymbol{v_{1i}}$$

$$\begin{bmatrix} v_{0i} \\ v_{1i} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{v0}^2 & \sigma_{v0v1} \\ \sigma_{v0v1} & \sigma_{v1}^2 \end{bmatrix} \right) \; or \; \begin{bmatrix} v_{0i} \\ v_{1i} \end{bmatrix} \sim N(0, \Sigma_v)$$


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Off Scale:

Empirical Bayes estimates of Subject Trends



Quick Question

- What does it mean to say that a variable related to our subjects is a time-varying covariate vs. a non-time-varying (aka, time-invariant) covariate?

Model with **time-invariant covariates**

- Within-subject (level-1) model:

$$Y_{ij} = b_{0i} + b_{1i}t_{ij} + \varepsilon_{ij}$$

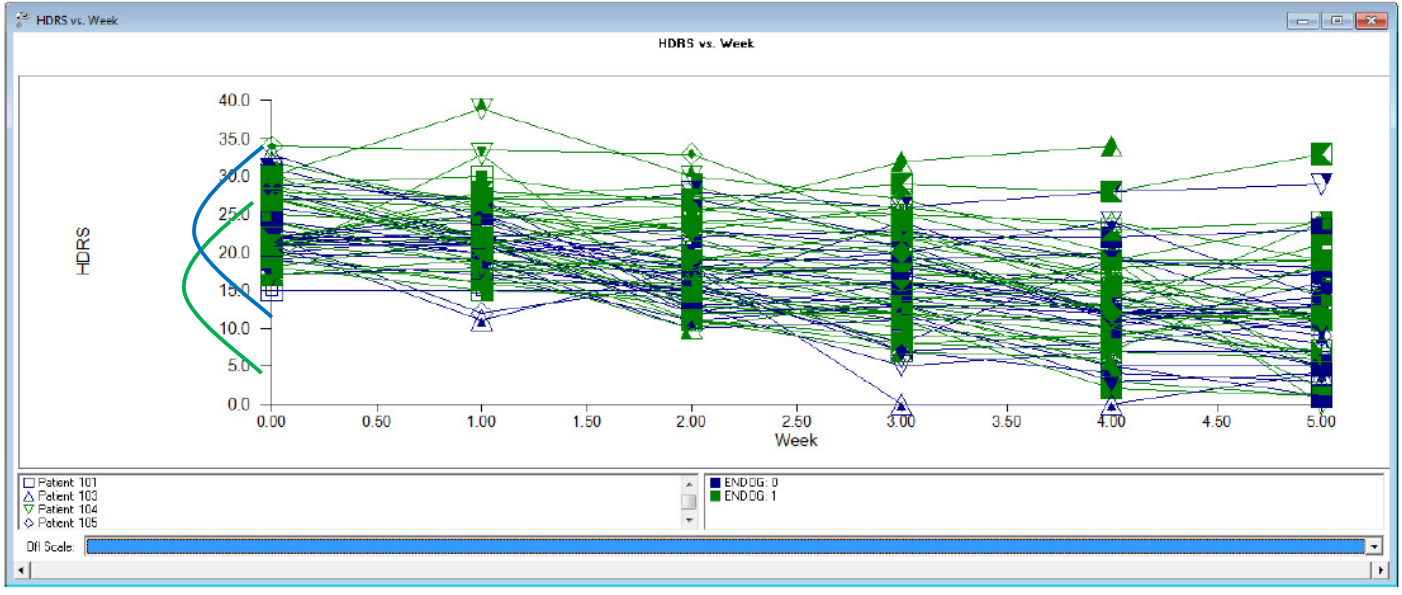
$$\varepsilon_{ij} \sim N\,(0, \sigma^2)$$

- Between-subject (level-2) model:

$$b_{0i} = \beta_{00} + \boldsymbol{\beta_{01}} \boldsymbol{G} + v_{0i}$$

$$b_{1i} = \beta_{10} + \boldsymbol{\beta_{11}} \boldsymbol{G} + v_{1i}$$

$$\begin{bmatrix} v_{0i} \\ v_{1i} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{v0}^2 & \sigma_{v0v1} \\ \sigma_{v0v1} & \sigma_{v1}^2 \end{bmatrix} \right) \; or \begin{bmatrix} v_{0i} \\ v_{1i} \end{bmatrix} \sim N(0, \Sigma_v)$$



- Plot of Endogenous and Non-Endogenous patients

Model with **time-varying covariates**

- Within-subject (level-1) model:

$$Y_{ij} = b_{0i} + b_{1i}t_{ij} + \boldsymbol{b_{2i}X_{ij}} + \varepsilon_{ij}$$

$$\varepsilon_{ij} \sim N\,(0, \sigma^2)$$

- Between-subject (level-2) model:

$$b_{0i} = \beta_0 + v_{0i}$$

$$b_{1i} = \beta_1 + v_{1i}$$

$$\boldsymbol{b_{2i} = \beta_2}$$

$$\begin{bmatrix} v_{0i} \\ v_{1i} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{v0}^2 & \sigma_{v0v1} \\ \sigma_{v0v1} & \sigma_{v1}^2 \end{bmatrix} \right) \; or \begin{bmatrix} v_{0i} \\ v_{1i} \end{bmatrix} \sim N(0, \Sigma_v)$$

Model with **time-varying covariates**

$$\boldsymbol{X_{ij}} = (\boldsymbol{X_{ij}} - \overline{\boldsymbol{X_i}}) + \overline{\boldsymbol{X_i}}$$

- Within-subject (level-1) model:

$$Y_{ij} = b_{0i} + b_{1i}t_{ij} + b_{2i}(\boldsymbol{X_{ij}} - \overline{\boldsymbol{X_i}}) + \varepsilon_{ij}$$

$$\varepsilon_{ij} \sim N\,(0, \sigma^2)$$

- Between-subject (level-2) model:

$$b_{0i} = \beta_0 + \boldsymbol{\beta_3}\overline{\boldsymbol{X_i}} + v_{0i}$$

$$b_{1i} = \beta_1 + v_{1i}$$

$$b_{2i} = \beta_2$$

$$\begin{bmatrix} v_{0i} \\ v_{1i} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{v0}^2 & \sigma_{v0v1} \\ \sigma_{v0v1} & \sigma_{v1}^2 \end{bmatrix}\right) \; or \; \begin{bmatrix} v_{0i} \\ v_{1i} \end{bmatrix} \sim N(0, \Sigma_v)$$

Model with **time-varying covariates**

Total effect of X:

$$\beta_2(\boldsymbol{X_{ij}} - \overline{\boldsymbol{X_i}}) + \boldsymbol{\beta_3}\overline{\boldsymbol{X_i}}$$

within-subject effects + between-subject effects

- How to test if the within-subject effect is equal to the between-subject effect?
  - $H0$: $\beta_2 = \beta_3$

Discussion

- Think about an example of a dataset that you could use MRMs for in your field (or related to another class you're taking, or that you've already worked with)?

OR

- What do you still feel confused about regarding MRMs?